Milestone Report: Financial Statement Fraud Detection under the SOX Regime Shift

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1 Revisions since the proposal

The domain question and intended use are unchanged: a pre-screening ranker that helps retail investors and forensic auditors decide which U.S. listed firms deserve forensic review, where a missed fraud costs far more than a false alarm. Three scope changes were made. (i) **Test window.** AAER enforcement follows a misstatement by several years, so recent labels are incomplete; labeled frauds in our panel fall from 86 firm-years in 2000 to 26 in 2008. The proposed 2016–2022 window is replaced by 2003–2008, the window used by Bao et al. [1]. (ii) **Staged features.** The milestone uses financial-statement variables only; 10-K text and the macro-regime feature move to the final report. (iii) **Adapted Beneish baseline.** SG&A is not in the data, so SGAI is fixed at its neutral value 1 and gross PPE replaces net PPE.

Current thesis. For retail investors and forensic auditors screening U.S. listed firms for accounting-fraud risk, we examine whether models trained on pre-SOX financial-statement data still rank post-SOX fraud firm-years better than the Beneish M-score and a Dechow-ratio logit, and whether 10-K text features recover the performance lost under the regime shift, using AAER-labeled firm-years from Bao et al. [1], with firm size as a second stress condition. The proposal's revision trigger "post-SOX AUC < 0.70" fired (Section 4), which is why the thesis now centers on out-of-time performance.

2 Data, pipeline, and baselines

Evidence. Firm-year panel of Bao et al. [1], fiscal years 1991–2008: 108,399 firm-years from 15,320 firms, of which 837 (0.77%) are labeled misstated based on SEC Accounting and Auditing Enforcement Releases (AAERs). Each row holds 28 raw Compustat variables and 14 ratios from Dechow et al. [3].

Pipeline (`fraud_milestone.py`). Rows are sorted by firm and year; one-year lags are used only when the prior fiscal year is consecutive, and divisions by zero become missing. Beneish components can be computed for 79–85% of firm-years; they are winsorized at the 1st/99th percentiles of the training split, and missing components are set to neutral values (1 for indices, 0 for TATA). Learned models impute with the training median.

Baselines. (1) *Beneish M-score* [2] with the published 8-variable coefficients, not re-fitted; we rank by the continuous score and also report the published cutoff $M > -1.78$. (2) *Dechow-ratio logit*: logistic regression on the 14 ratios, re-estimated on the training split, following the benchmark design of Bao et al. [1]. **Models.** Logistic regression on the 28 raw variables; random forest (300 trees, min. leaf 5) and histogram gradient boosting (300 iterations, learning rate 0.05) on raw variables plus ratios. All learners use balanced class weights; nothing is tuned yet.

Splits. *Random, firm-grouped* (optimistic reference): 75/25 by firm, so no firm is in both sets; 80,909 training firm-years (589 frauds), 27,490 test (248). *SOX out-of-time*: train 1991–2000 (60,741; 355 frauds), test 2003–2008 (35,166; 261 frauds). Years 2001–2002 are excluded as the SOX transition

Table 1: Test-set performance. AUC with 95% bootstrap CI (500 resamples); P@1% and R@1% = precision and recall among the top 1% highest-scored firm-years.

Split	Model	AUC [95% CI]	P@1%	R@1%
Random (firm-grouped)	Beneish M-score (fixed)	0.544 [0.508, 0.584]	0.007	0.008
	Dechow-ratio logit	0.708 [0.681, 0.738]	0.015	0.016
	Logit (28 raw)	0.744 [0.711, 0.774]	0.076	0.085
	Random Forest	0.758 [0.730, 0.785]	0.069	0.077
	Gradient Boosting	0.782 [0.759, 0.807]	0.055	0.060
	base rate		0.0090	
SOX out-of-time	Beneish M-score (fixed)	0.554 [0.522, 0.585]	0.009	0.011
	Dechow-ratio logit	0.658 [0.630, 0.687]	0.003	0.004
	Logit (28 raw)	0.654 [0.621, 0.687]	0.011	0.015
	Random Forest	0.650 [0.619, 0.686]	0.009	0.011
	Gradient Boosting	0.647 [0.615, 0.682]	0.009	0.011
	base rate		0.0074	

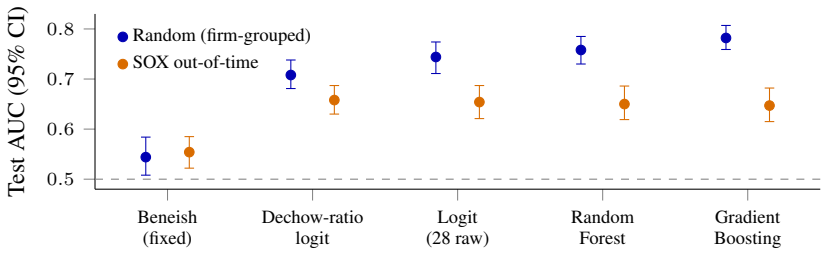


Figure 1: Test AUC with 95% bootstrap intervals by model and split. Dashed line: random ranking.

and as a two-year detection gap. 63 pre-SOX fraud firm-years belonging to schemes that continue into the test period were dropped from training. **Measures.** AUC with a 95% bootstrap interval (500 resamples), and precision and recall among the top 1% of scored firm-years (P@1%, R@1%), matching the decision of sending a short list to forensic review. Accuracy is not reported because the base rate is below 1%.

3 Preliminary results

O1: Learned models beat the fixed formula, also after SOX. On the SOX split every learned model reaches AUC 0.647–0.658, versus 0.554 [0.522, 0.585] for Beneish; the intervals do not overlap (Table 1, Figure 1). At its published cutoff, Beneish flags 15.4% of post-SOX firm-years and catches 44 of 261 frauds; flagging the same share at random would catch about 40.

O2: The ML advantage does not survive the time split. On the random split, gradient boosting (0.782 [0.759, 0.807]) beats the Dechow-ratio logit (0.708 [0.681, 0.738]). On the SOX split, the four learned models are within 0.011 AUC of each other with overlapping intervals. The drop from the random to the SOX split grows with model flexibility: -0.050 (Dechow logit), -0.090 (raw logit), -0.108 (random forest), -0.135 (gradient boosting); the unfitted Beneish score does not drop ($+0.010$).

O3: The top of the ranking fails out of time. On the random split, the raw logit's top 1% contains 21 frauds among 275 firm-years (P@1% = 0.076, 8.5$\times$ the base rate). On the SOX split, every model's top 1% holds 1–4 frauds among 352, while random selection would be expected to hold 2.6. With so few hits the estimate is noisy, but it gives no evidence that the shortlist works after SOX.

O4: Firm size (HC-3, started). Frauds concentrate in large firms: 153 of the 261 post-SOX frauds

are in the largest total-asset tercile and 35 in the smallest. Every model ranks worst inside the large tercile (e.g., Beneish 0.537 [0.487, 0.581], an interval that includes 0.5). For the raw-variable logit, the pooled AUC (0.654) exceeds its AUC in every tercile (0.55–0.61), so part of its ranking separates large from small firms rather than fraud from non-fraud within a size class; raw dollar amounts encode scale. The scale-free Dechow ratios do best in small firms (0.729 [0.657, 0.789], only 35 frauds).

Limited attempts. Our Beneish score is a lower-fidelity version (one component fixed, a PPE proxy, 15–21% of components filled), and the original was validated on an industry-matched sample, not a full population at a 0.7% base rate, so its low AUC is not a verdict on the original study.

4 What the evidence supports so far

Supported (one split, untuned models). Models re-estimated on data rank fraud about 0.10 AUC better than the fixed 1999 formula even when trained only on pre-SOX years, but flexible ML adds nothing over a linear ratio model out of time, and no model reaches the proposal's 0.70 post-SOX threshold. Under the proposal's own trigger, the ranking claim is bounded to the training regime, and the shortlist use is not yet supported after SOX. **Unresolved.** (1) Whether the drop is caused by SOX or by any multi-year gap: the random split mixes periods and includes the 2000–2002 fraud peak. (2) How much of the raw-variable models' signal is firm size. (3) Label noise: undetected frauds count as clean, and detection lag affects later test years more (69 labeled frauds in 2003 versus 26 in 2008). (4) Whether 10-K text recovers the lost performance.

5 Class-survey case summary and matrix entry

Case summary. Accounting-fraud screening for U.S. listed firms, where current practice is a fixed 1999 formula. Using AAER-labeled firm-years, we test whether models trained before the 2002 Sarbanes-Oxley Act still rank fraud afterwards.

Domain	Intended use	Baseline	Evidence	Shift tested	Current finding
Forensic accounting	Rank firms for fraud review	Beneish M-score; Dechow-ratio logit	AAER labels + Compustat (Bao et al.)	Pre- $\rightarrow$ post-SOX; firm size	Post-SOX AUC 0.65 vs. 0.55 for Beneish; ML = linear model; top-1% precision at base rate

6 Next stress conditions and completion plan

Weeks 1–2 (stress tests on the current pipeline). *Placebo break*: train 1991–1995, test 1998–2000 with the same gap; a similar drop would point to generic time decay rather than SOX. *Rolling yearly test*: train on years up to $t-2$, test on t, to measure decay per year of gap. *Size control*: scale raw variables by total assets, add log assets, report within-tercile AUC for all models. Add P@5% and NDCG with bootstrap intervals, and tune on a time-ordered validation set. **Weeks 3–4 (text).** Link `gvkey` to CIK and SIC through WRDS; extract MD&A from EDGAR for test-period firms; add Loughran–McDonald tone and year-over-year similarity; ablate financial-only versus financial + text on the SOX split. **Week 5.** Cross-industry test (HC-2, using SIC), calibration curve, final write-up. **Risks.** If WRDS access fails, HC-2 is replaced by the rolling test. If MD&A extraction is not done by week 4, the final report keeps the financial-statement result and reports text features as a pilot on a subsample.

Contributions. Jiashuo Xu (sole author): all data processing, modeling, evaluation, and writing. **Code:** `https://github.com/prcsurexu/fraud-milestone`

References

[1] Y. Bao, B. Ke, B. Li, Y. J. Yu, and J. Zhang. Detecting accounting fraud in publicly traded U.S. firms using a machine learning approach. *Journal of Accounting Research*, 58(1):199–235, 2020.

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[3] P. M. Dechow, W. Ge, C. R. Larson, and R. G. Sloan. Predicting material accounting misstatements. *Contemporary Accounting Research*, 28(1):17–82, 2011.